


Data Science Program

Capstone Report - Spring 2026

# Predicting Global Cyberlaw Adoption

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*A capstone presented in fulfillment of the requirements for the degree of Bachelor's of Science in Data Science*

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## Preface

This project examines global patterns in cyberlaw adoption and the factors that shape legislative readiness across countries. My background in data analysis and interest in digital governance motivated this study, especially as cyberlaw has become increasingly important in a rapidly digitalizing world. The report is intended for readers interested in international digital policy, comparative governance, and the measurement of legal development. By combining descriptive visualization with cross-national modeling, the project aims to provide a clearer and more accessible view of how countries differ in their approach to digital legislation.


## Acknowledgements

I would like to thank the instructors and guiding staff who guided this project, as well as classmates who offered helpful feedback during the development of the analysis. I am also grateful for the support of friends and family, whose encouragement made this work possible.

**Abstract**

*This project examines why countries differ in their adoption of cyberlaw frameworks. The question is important because digital legislation has expanded rapidly, yet global patterns remain uneven and difficult to explain. Using descriptive visualizations and cross national regression models, the study evaluates the influence of development, governance, and digital economic activity on cyberlaw readiness. The results show that development level is the strongest and most consistent predictor, ICT Goods Exports have a smaller but positive effect, and democracy plays little role once other factors are considered. A comparison with a harmonized 2019 model indicates that these patterns remain broadly stable over time. The findings suggest that structural capacity, rather than political system type, shapes how countries build digital legislation.*

**Keywords**

**Cyberlaw; Digital Legislation; Cross-National Analysis; Human Development; ICT Exports; Governance; Post-COVID Comparison; Capstone; data science; GWU**

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# 1 Reproducibility and Code Availability

All data cleaning, visualization, and modeling code used in this report is fully reproducible and available at:

`https://github.com/GW-dataset/26-spring-FU-U/tree/main`

The repository includes the complete analysis pipeline, environment information, and instructions for regenerating all figures and results.

# 2 Introduction

## 2.1 General Guide in

In historical development patterns, many "catch-up" countries experience an early phase where rapid industrialization and technological adoption come with relatively low emphasis on privacy protection. During this stage, copying, imitation, and rapid diffusion of technology often take precedence over data regulation. However, once these countries reach a development level comparable to that of top-tier economies, they tend to shift toward stronger privacy norms and legal protections.

The objective of this project is to explain why countries differ in their adoption of key digital governance legislation by analyzing cross national variation in regulatory capacity. Using UNCTAD's global dataset on Electronic Transactions, Consumer Protection, Privacy and Data Protection, Cybercrime, and Indirect Taxation, the project evaluates how political, institutional, and developmental factors shape the likelihood that a country implements comprehensive digital rights frameworks. The analysis aims to identify the structural conditions that best predict strong legislative adoption and to clarify the broader patterns underlying global disparities in digital governance. The final results show that overall human development and institutional maturity are the strongest predictors of comprehensive digital governance adoption, aligning closely with the project's initial theoretical expectations.

## 2.2 Research Questions (SMART)

This study addresses the following SMART research questions:

1. To what extent does a country's Human Development Index (HDI) predict its Cyberlaw Legislation Index score in 2026?

2. Has the influence of HDI, democracy, and ICT exports on cyberlaw adoption changed between 2019 and the post-COVID period?

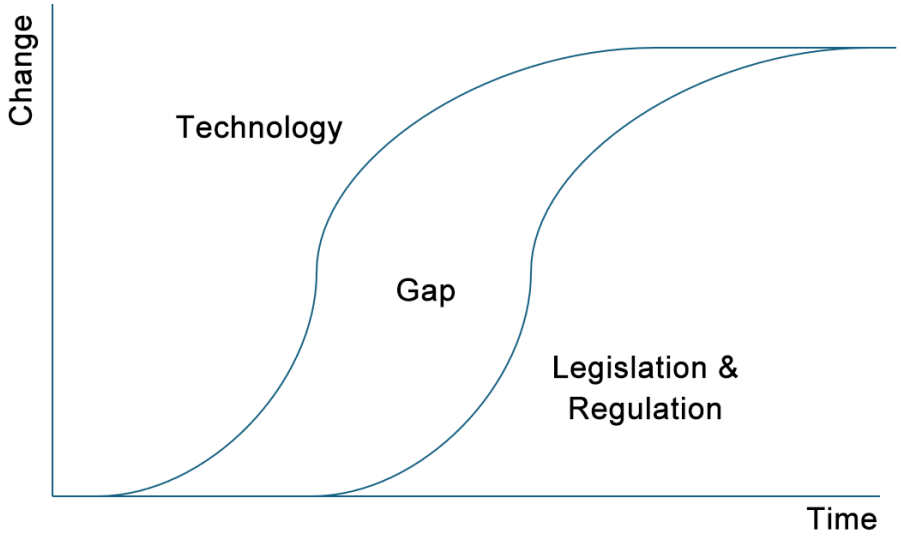



Figure 1: Conceptual model illustrating the widening gap between technological acceleration and regulatory development.

### 2.3 Clarification for non-goals

This project focuses on explaining cross-national variation in Cyberlaw adoption using publicly available structural indicators. It does not evaluate the legal quality, enforcement capacity, or effectiveness of specific laws, nor does it attempt to establish causal relationships between predictors and legislative outcomes. The analysis also does not forecast future cyberlaw adoption or assess the political motivations behind individual countries' policy choices.

### 2.4 Structure Preview

The remainder of this report is structured as follows. Section 3 reviews related work and situates this project within existing research. Section 4 describes the methodological approach, including raw data visualization, data preparation, model specification, and assumption checks. Section 5 presents the empirical results, including initial visualizations, model performance, predictor effects, and a comparison with the 2019 baseline model. Section 6 provides a broader discussion of the findings, while Section 7 outlines limitations and directions for future work. Section 8 concludes the report, and Section 9 addresses ethical considerations.

## 3 Related Work

### 3.1 Works

1. Study1: This paper [1] breaks down the major privacy and security risks that come with generative AI, showing how models can leak data, be manipulated, or produce harmful content. It also reviews

the current defenses and argues that we need clearer rules, better transparency, and stronger safeguards as these systems keep spreading. It is an expansion of focus toward a more recent topic. This study focuses on technical risks and safeguards at the system level, whereas my project examines how national structural conditions shape legislative responses to such risks, extending the discussion from micro-technical vulnerabilities to macro-regulatory capacity.

2. Study2: This study [2] examines how cultural differences shape privacy decisions on social network sites across the US, Korea, and China, revealing that individualistic and collectivistic cultures prioritize different contextual factors and privacy concerns when managing friend requests, information sharing, and perceived risks. While this study analyzes individual privacy decisions shaped by cultural context, my project shifts the focus to country-level institutional and developmental factors, connecting micro-level privacy behavior to macro-level legislative outcomes.

3. Study3: This study [3] explores Generation Z's attitudes (I want to see what attitude does the new generation who are or would be the new labor source to society) toward artificial intelligence, revealing a mix of optimism about AI's potential and concern over privacy and ethical risks. Based on focus groups across nationalities, it highlights trust as a key factor shaping Gen Z's relationship with emerging technologies. This study highlights generational attitudes toward AI, whereas my project investigates how national structural capacity—not demographic attitudes—predicts the adoption of digital governance laws, situating these perceptions within broader institutional patterns.

4. Study4: In the article [4], the authors look into the statistical test to verify the difference in privacy awareness for different age group, the findings are that the young adults and older adults are similar; both have privacy concerns but often lack knowledge to protect themselves, while the middle-aged adults tend to show higher awareness and knowledge. Although this study compares privacy awareness across age groups, my project extends beyond individual knowledge gaps to examine how national development and governance structures influence the formation of comprehensive digital legislation.

5. Study5: This study [5] investigates how security, individuality, and reputation influence Vietnamese consumers' trust, perceived risk, and attitudes toward online shopping. Using SEM analysis on survey responses, the authors find that security and reputation boost cognitive trust and reduce perceived risk, which in turn shape positive attitudes and increase purchase intention. (A parallel of my study, but in a more specific level) More specifically, this study operates at the consumer-behavior level, whereas my project analyzes structural national predictors of legislative adoption, linking micro-level trust dynamics to macro-level governance capacity.

6. Study6: Similar to the the previous article [5], this study [6] compares generational attitudes toward state and corporate surveillance across Estonia, Portugal, and Sweden, revealing that older indi-

viduals tend to tolerate state surveillance due to institutional trust, while younger individuals are more accepting of corporate dataveillance, shaped by confident media use and global techno-cultural influences. While this study compares generational attitudes toward surveillance across countries, my project focuses on the structural determinants of national legislation, moving from attitudinal tolerance to institutional capacity as the primary explanatory factor.

7. Study7: This study [7] examines how data has become a distinct form of capital within contemporary political-economic systems. Sadowski argues that datafication is driven by cycles of accumulation and extraction, where organizations seek to capture data from all possible sources with minimal regard for consent or compensation. Although not focused on national legislation, the study highlights the structural pressures that push states and institutions toward stronger governance frameworks. This provides a conceptual foundation for understanding why digital-governance laws emerge unevenly across countries, complementing this project's emphasis on structural and developmental predictors. Building on this structural perspective, my project empirically tests how developmental and institutional factors translate these pressures into measurable differences in national digital-governance laws.

8. Study8: This study [8] presents the updated Correlates of War Intergovernmental Organizations (IGO) dataset, offering comprehensive country-year and organization-year membership data from 1816 to 2014. Pevehouse, Nordstrom, and Jamison show how IGO participation reflects states' institutional engagement, international integration, and governance capacity. While not directly about digital legislation, the dataset's insights into institutional connectivity and global embeddedness support this project's argument that countries with stronger institutional linkages and administrative capacity are more likely to adopt comprehensive digital-governance frameworks. My project extends this logic by demonstrating how such institutional embeddedness predicts concrete legislative outcomes in the domain of digital governance.

## 3.2 General note

Overall, the existing research falls into two main areas: studies that look at people's privacy attitudes, risk perceptions, and generational differences, and studies that discuss how digitalization shapes broader governance systems. The micro-level work offers detailed insights into how individuals think about and respond to digital risks, while the more structural work highlights how social, economic, and political conditions influence national approaches to data governance. However, these two strands rarely connect. Micro-level studies do not usually extend their findings to national policy outcomes, and structural studies often lack cross-national, quantitative testing. This project brings the two together by turning individual-level mechanisms into measurable country-level factors and examining how development, institutional

capacity, and digital economic engagement shape the adoption of cyber-related laws across countries.

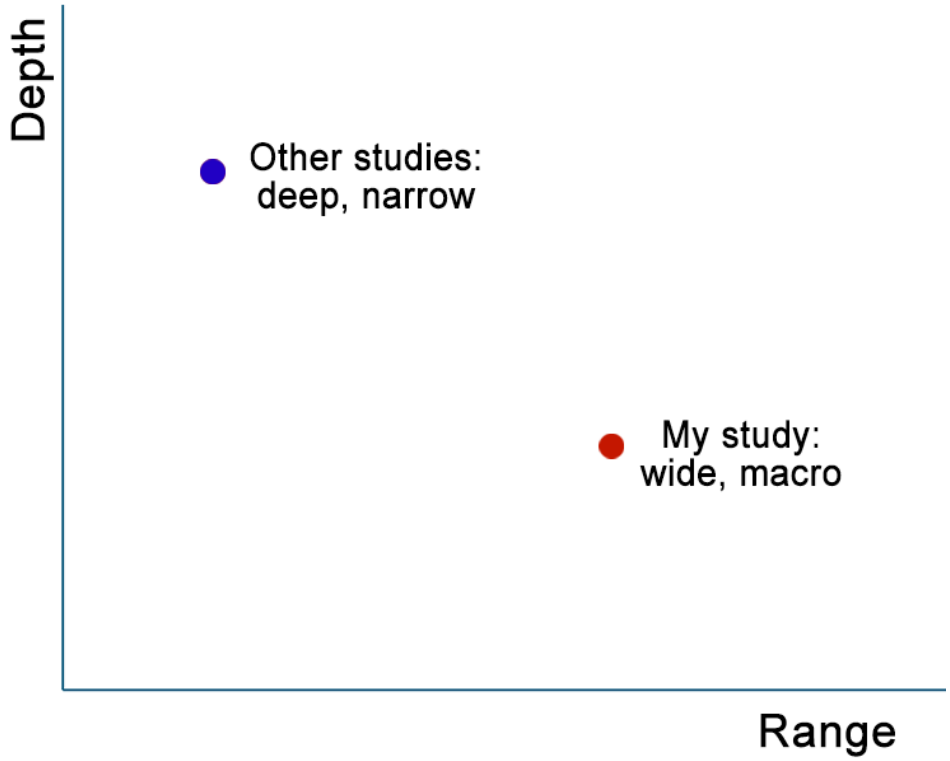



Figure 2: Conceptual illustration of the difference of my study and the other studies in the field

## 4 Method

This study examines cross-national predictors of cyberlaw development using publicly available datasets from the United Nations Conference on Trade and Development (UNCTAD), the United Nations Development Programme (UNDP), and the Economist Intelligence Unit (EIU). All variables use the most recent available data: Human Development Index (HDI) (2023) [9], Democracy Index (2024) [10], Information and Communication Technology (ICT) Goods Exports (2023) [11], and Cyberlaw Legislation Index (2026) [12]. The initial dataset included 195 countries before handling missing values. And the final analytic sample: N = 138 countries. The Cyberlaw Legislation Index is constructed from five categories of legislation recorded by UNCTAD (Electronic Transactions, Consumer Protection, Privacy and Data Protection, Cybercrime, and Indirect Taxation), which are treated equally when accounting to the index–No legislation = 0; Draft/Have a legislation = 1, and together, this builds a 0 5 index of Cyberlaw legislation.

## 4.1 Raw Data Visualization

Before estimating the regression models, the Cyberlaw Index was visually examined to understand the global distribution of digital legislation. Two descriptive visualizations were produced to summarize the raw patterns in the dependent variable. The first visualization compares the number of countries with legislation across the five UNCTAD categories—Electronic Transactions, Consumer Protection, Privacy and Data Protection, Cybercrime, and Indirect Taxation—highlighting substantial variation in adoption rates across legal domains. The second visualization maps the Cyberlaw Index globally, illustrating broad regional differences in legislative readiness and revealing clusters of high- and low-scoring countries. The figures themselves are presented in the Results section.

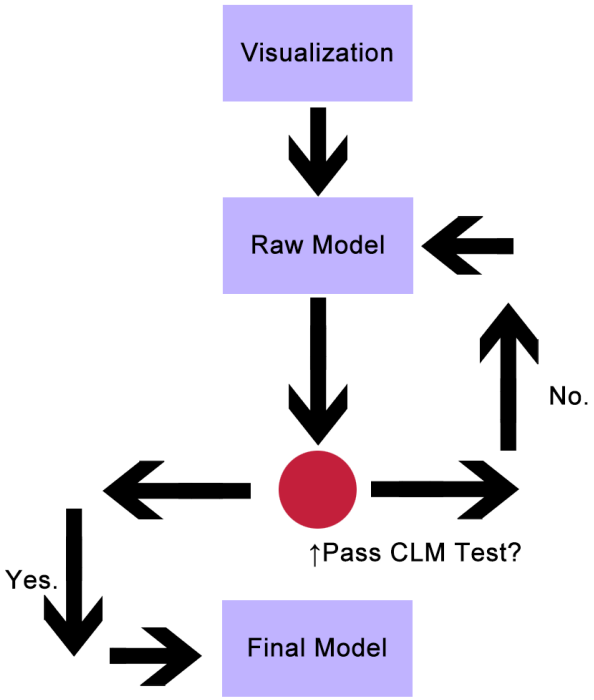



Figure 3: Process map

## 4.2 Data Preparation

The dependent variable is the Cyberlaw Index, which measures national adoption of digital legislation. Predictor variables include the Human Development Index (HDI), Democracy Index, and ICT Goods Exports. HDI was selected as a composite measure of education, life expectancy, and income, while ICT exports capture digital economic activity and the Democracy Index reflects governance quality.

All datasets were accessed between February and April 2026 and permit academic reuse under their respective public licensing terms.

All numeric predictors were examined for skewness, outliers, and multicollinearity. ICT Goods Exports and GNI showed left-skewed distributions, so log transformations were applied to improve normality.

Variance Inflation Factors (VIFs) were calculated for all predictors to assess potential multicollinearity. These diagnostics informed the variable selection process described in Section 3.3.

The Dataset Schema and Missingness Summary are shown in Table 1 below.

Table 1: Dataset Schema and Missingness Summary

| Variable | Type | Units | Description | Missing |
|---|---|---|---|---|
| HDI_2019 | Numeric | Index (0–1) | UNDP Human Development Index for 2019 | 10 |
| HDI_2023 | Numeric | Index (0–1) | UNDP Human Development Index for 2023 | 18 |
| DemIdx_2019 | Numeric | Score (0–10) | EIU Democracy Index for 2019 | 34 |
| DemIdx_2024 | Numeric | Score (0–10) | EIU Democracy Index for 2024 | 63 |
| ICTExp_2019 | Numeric | Million USD | ICT goods export value (current prices) for 2019 | 14 |
| ICTExp_2023 | Numeric | Million USD | ICT goods export value (current prices) for 2023 | 41 |
| Idx5_2026 | Numeric | Index (0–5) | UNCTAD Cyberlaw Legislation Index (five-category) for 2026 | 0 |
| Idx3_2019 | Numeric | Index (0–3) | Harmonized Cyberlaw Index (three-category) for 2019 | 0 |

## 4.3 Model Specification

Multiple linear regression models were estimated to identify predictors of Cyberlaw Index scores. The modeling process began with a full set of available numeric variables (HDI components, GNI, schooling indicators, life expectancy, ICT exports, Democracy Index). Variables with severe multicollinearity or conceptual redundancy were removed, resulting in a final model including HDI, Democracy Index, and log ICT Goods Exports.

Parallel models were constructed using 2019 data (The 2019 model serves as a baseline for comparison) to compare pre- and post-COVID predictor influence. Difference-in-Differences and Chow tests were considered but rejected due to the cross-sectional structure of the dataset. Instead, standardized coefficients were used for effect-size comparison.

## 4.4 Assumption Checks

Classical Linear Model (CLM) assumptions were evaluated using standard diagnostic plots and statistical tests. Linearity was assessed through residual versus fitted and component plus residual plots, which showed no systematic patterns. Homoscedasticity was examined using the Scale–Location plot, indicating relatively constant variance across fitted values. Normality of residuals was evaluated with a Q–Q plot, which showed only minor deviations at the tails. Independence of errors was confirmed by the Durbin–Watson test (DW = 2.03, p = 0.83), suggesting no autocorrelation. Overall, the diagnostics indicate that the model satisfies the key CLM assumptions, supporting the validity of subsequent regression estimates. Full diagnostic plots are provided below.

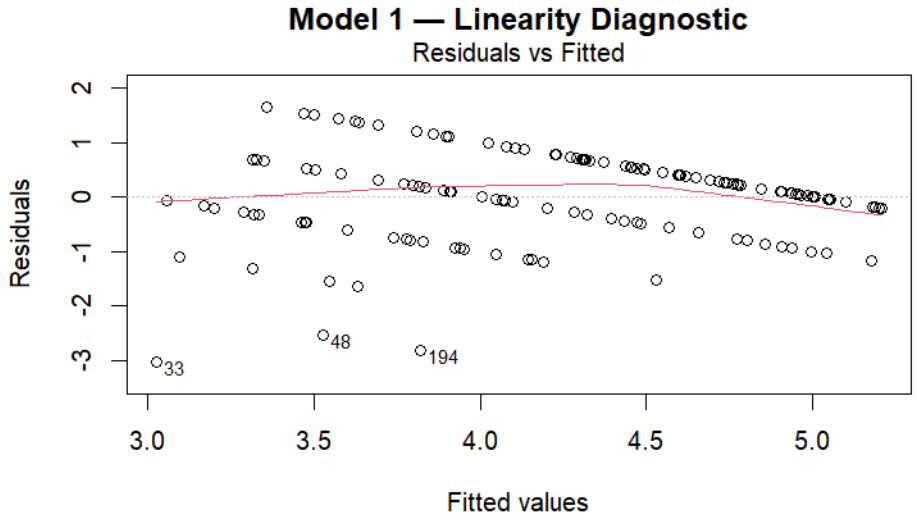



Figure 4: Residuals vs Fitted plot assessing linearity and potential model misspecification.

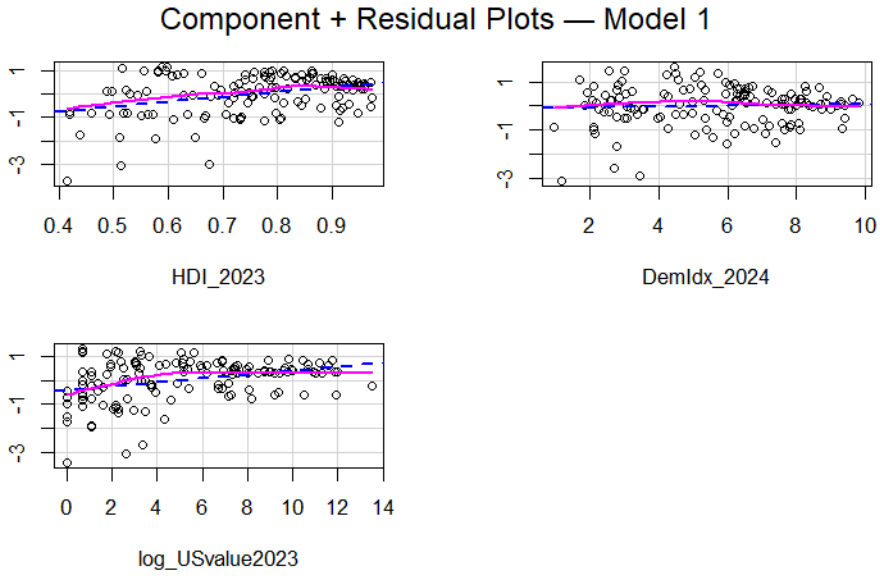



Figure 5: Residuals vs Leverage plot identifying influential observations.

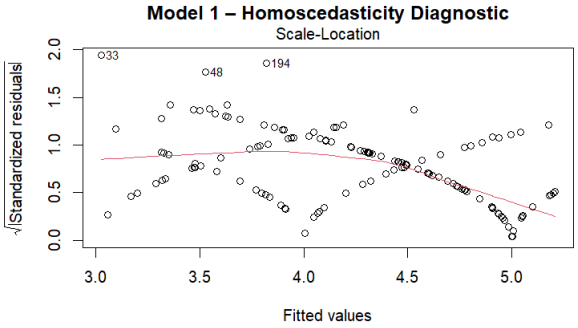



Figure 6: Scale–Location plot examining homoscedasticity across fitted values.

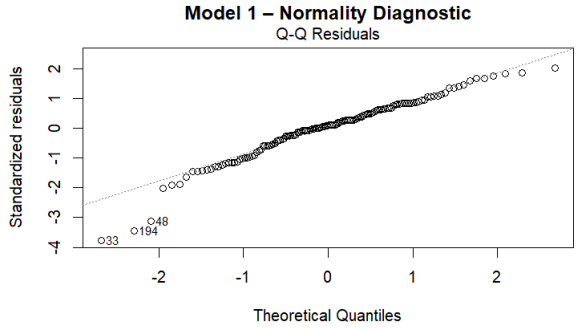



Figure 7: Normal Q–Q plot evaluating the normality of residuals.

# 5 Results and Discussion

## 5.1 Initial Visualization

As introduced in Section 3.1, the preliminary visualizations provide a descriptive overview of global cyberlaw development. The bar chart shows substantial variation across legislative domains, with Cybercrime and Electronic Transactions exhibiting the highest adoption levels, while Indirect Taxation remains the least implemented. The world map further highlights pronounced regional disparities: countries in Europe and parts of Asia tend to score higher on the Cyberlaw Index, whereas many states in Africa and Latin America display lower levels of legislative readiness. These patterns contextualize the regression results presented in the following sections.

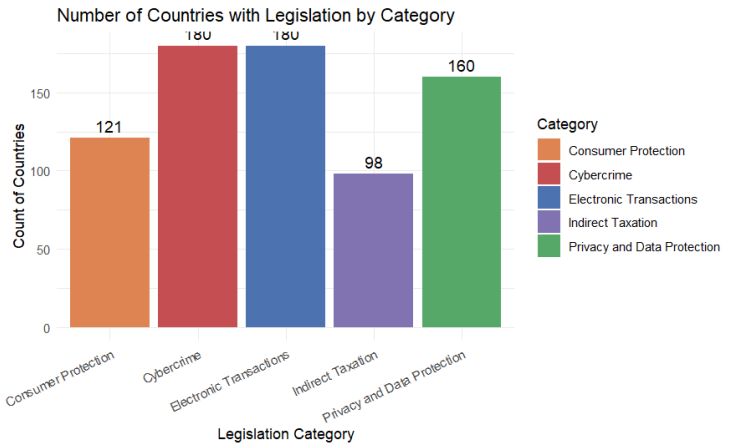



Figure 8: Barchart visualization of country count for each category of legislation.

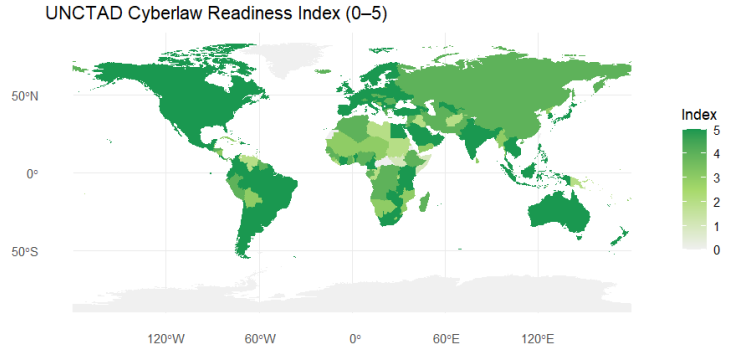



Figure 9: Map visualization of 0 5 Cyberlaw index.

## 5.2 Model Overview

$$\text{Legislation Index} = 2.138478 + 2.079351\,(\text{HDI}) + 0.020514\,(\text{Democracy Score}) + 0.079298\,(\text{ICTExports}) \quad (1)$$

The final regression model for 2026 includes three predictors: the Human Development Index (HDI), the Democracy Index, and the log-transformed value of ICT Goods Exports. These variables were retained after removing predictors with severe multicollinearity or conceptual overlap with HDI. The resulting specification provides a concise and theoretically grounded framework for explaining cross-national variation in Cyberlaw adoption.

## 5.3 Predictor Effects (2026 Model)

The Human Development Index (HDI) is the strongest predictor of Cyberlaw Index scores (b = 2.079, p = 0.0076). Countries with higher levels of education, life expectancy, and income tend to adopt substantially more comprehensive digital legislation. ICT Goods Exports also show a statistically significant positive association with cyberlaw adoption (b = 0.079, p = 0.0052), indicating that digitally engaged economies

are more likely to implement cyber-related legal frameworks. In contrast, the Democracy Index does not exhibit a statistically significant effect (b = 0.020, p = 0.608). This suggests that, after accounting for developmental capacity and digital economic activity, governance structure does not meaningfully predict cyberlaw readiness in 2026.

### 5.4 Model Fit

The model explains a moderate portion of global variation in Cyberlaw adoption $R^2 = 0.3375$, Adjusted $R^2 = 0.3227$ ($R^2$ and standardized coefficients were used because the goal is explanatory modeling rather than prediction.). This indicates that roughly one-third of cross-national differences in Cyberlaw Index scores can be accounted for by developmental capacity, governance, and digital economic activity. Combined with the strong diagnostic performance reported in Section 3.4, these results suggest that the model is well-specified. Among the predictors, HDI contributes the largest share of explained variance, while ICT Goods Exports provide additional explanatory power and the Democracy Index does not meaningfully improve model fit.

### 5.5 Comparison with 2019 Model

To compare cyberlaw readiness before and after the COVID-19 pandemic, the Cyberlaw Index must be harmonized across years. The 2026 index ranges from 0–5, reflecting five legislative categories, whereas the 2019 dataset only includes three categories, producing a 0–3 scale. To ensure comparability, the post-COVID model was re-estimated using the same three-category structure as in 2019. This adjustment avoids artificial inflation of post-COVID scores and ensures that differences reflect substantive legal changes rather than differences in index construction.

Table 2 below summarizes the standardized effects of HDI, the Democracy Index, and ICT Goods Exports on cyberlaw adoption in 2019 and in the post-COVID period. These coefficients allow direct comparison of predictor importance across years.

| Standardized Coefficients: 2019 vs Post-COVID | | |
|---|---|---|
| Predictor | 2019 Model ($\beta$) | Post-COVID Model ($\beta$) |
| HDI | 0.3354 | 0.2889 |
| Democracy Index | 0.1673 | 0.0822 |
| ICT Goods Exports (log) | 0.0439 | 0.0370 |

Table 2: Comparison of standardized predictor effects before and after COVID-19.

## 6 Discussion

The results show that national development remains the most consistent factor behind cyberlaw adoption. Countries with higher levels of education, income, and life expectancy tend to have more complete digital

legislation, both before and after the COVID 19 pandemic. This pattern suggests that basic institutional and human capacity continues to shape how quickly countries can build and maintain cyberlaw frameworks. ICT Goods Exports also show a positive association with cyberlaw adoption, but their influence becomes smaller once the index is reduced from five categories to three. This drop is likely related to the removal of Consumer Protection and Indirect Taxation from the post COVID comparison model, as both categories may be more closely connected to trade and digital market activity. Without these components, the link between ICT exports and overall cyberlaw readiness becomes weaker, though it remains statistically meaningful. The Democracy Index, however, does not show a meaningful effect in the post COVID model. While it had a moderate association in 2019, its influence weakens once development and digital economic activity are taken into account. This suggests that political system type is not a reliable predictor of cyberlaw readiness. Many non democratic or hybrid regimes appear to adopt cyber legislation for practical reasons—such as economic competitiveness or security—rather than because of democratic institutions. Comparing the two models also shows a slight shift in the relative importance of the predictors. HDI remains the strongest factor, but its influence decreases somewhat after COVID 19, likely because more countries have adopted at least some cyberlaw components, reducing the overall variation. Meanwhile, the decline in the democracy effect and the stable contribution of ICT exports point toward a more uniform global trend: cyberlaw adoption is becoming driven less by political values and more by capacity and economic needs. Overall, the findings suggest that cyberlaw development is shaped primarily by structural conditions rather than political regime type. Countries with stronger development foundations and deeper engagement in digital trade are better positioned to expand their legal frameworks, regardless of their political system. This shift has implications for international cooperation, as it indicates that progress in cyberlaw harmonization may continue even without broader political convergence.

# 7 Limitations and future works

## 7.1 Data Availability and Missingness

A major limitation of this study is the uneven availability of key indicators. Several countries lack reliable ICT export data or Democracy Index scores, and the missingness is not random. States with weaker statistical capacity or limited institutional transparency are less likely to report these measures, which reduces sample size and may bias the results toward more developed countries. This constraint limits the generalizability of the findings, especially for low-income regions.

### 7.2 Temporal and Structural Inconsistencies

The analysis combines indicators drawn from slightly different years, and the Cyberlaw Index itself is not fully consistent across time. The 2026 index includes five legislative categories, while the 2019 dataset contains only three. To enable comparison, the post COVID model was re estimated using the 0–3 structure, but this adjustment inevitably removes information—particularly in Consumer Protection and Indirect Taxation—that may be relevant for understanding the role of digital trade. As a result, some shifts in predictor importance may reflect index structure rather than substantive legal change.

### 7.3 Measurement Constraints

Cyberlaw development is complex and multidimensional, yet the index simplifies it into a small number of categories. This approach captures broad patterns but cannot reflect differences in legal quality, enforcement capacity, or institutional design. Similarly, HDI, ICT exports, and the Democracy Index represent only partial aspects of development, economic engagement, and governance. These measures provide useful proxies but do not fully capture the underlying processes that shape cyberlaw adoption.

### 7.4 Future Work

Future research could expand this analysis in several ways. First, building a longer historical dataset of cyberlaw adoption would allow for more dynamic modeling, such as identifying when legislative growth accelerates or how earlier development levels shape later legal outcomes. Second, improving data completeness—particularly for ICT indicators, democracy measures, and institutional capacity—would reduce missingness and strengthen cross-national comparisons. Finally, future work could examine additional structural or political factors, such as regime type, international agreements, or regional diffusion patterns, to capture influences not fully reflected in economic and developmental indicators. These extensions would provide a more comprehensive understanding of how and why cyberlaw frameworks evolve across countries.

## 8 Conclusion

This study set out to understand what drives differences in cyberlaw adoption across countries. The results show a clear pattern: countries with stronger development levels tend to have more complete digital legislation, and this holds both before and after the COVID 19 pandemic. ICT Goods Exports also play a role, though the effect becomes smaller when the index is reduced to three categories, likely because some trade related components are no longer included. The Democracy Index, however, does not show a meaningful influence once development and digital economic activity are considered. The comparison between 2019 and the post COVID period suggests that the overall structure of global cyberlaw adoption

has remained fairly stable, even as more countries have begun implementing basic legislation. While the specific strength of each predictor shifts slightly, the broader picture remains the same: development capacity matters most, and political system type matters least. Overall, the findings point to a practical rather than ideological pattern of cyberlaw growth. Countries with stronger institutions and greater involvement in the digital economy are better positioned to expand their legal frameworks, regardless of regime type. Although data gaps and index differences limit the precision of the analysis, the study provides a clearer view of the global landscape and highlights areas where future research can build a more complete understanding of how digital legislation evolves.

# 9 Ethical Considerations

All datasets used in this project (UNCTAD Cyberlaw Tracker, UNDP HDI, V-Dem democracy scores, and UN Comtrade ICT exports) are publicly available, aggregated at the country level, and contain no personally identifiable information. Licensing terms permit academic reuse, and no IRB review is required. The 2019 Cyberlaw Tracker data were obtained through an archived snapshot of the UNCTAD website. While this method is commonly used to recover discontinued public data, it introduces the possibility of version inconsistencies or missing entries. To mitigate this, multiple archived captures were cross-checked to ensure stability and accuracy.

Several dataset-specific biases were considered. HDI and democracy scores may embed structural or expert-coding biases, and ICT export data may undercount activity in countries with weaker reporting systems. To avoid unfair conclusions, the analysis focuses on structural patterns rather than normative judgments and avoids causal claims.

Privacy risks are low because all data are aggregated at the national level. Even so, the project avoids combining indicators that could create misleading or sensitive interpretations. For example, crime and surveillance variables were excluded to prevent unnecessary stigma or misinterpretation. Potential dual-use risks include misuse of results to pressure or rank countries. To reduce this, the report emphasizes capacity-based explanations rather than political evaluations.

The GAI Tool (ChatGPT-4) was used only for writing support (editing, restructuring, clarity). All statistical analysis, modeling decisions, and interpretations were performed manually. AI outputs were verified against the actual regression results to avoid hallucination, attribution drift, or over-reliance.

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